

Hoda Anvari-Kazemabad

Profile

Software engineer with experience in backend systems, research infrastructure, and industrial IoT. Focus on API design (REST/OpenAPI), authentication and access control, databases and CI/CD, with a research background in explainable machine learning for protein engineering and peptide drug discovery.

Professional Experience

Sep 2025 – Present **Software Engineer**, *Leibniz Institute of Plant Biochemistry (IPB)*, Halle (Saale)

- **CRIMSy & N1H (Natural One Health):** Designed and modernised the backend of the Natural One Health research infrastructure, linking natural product chemistry, biodiversity, and biomedical research data across distributed nodes.
- **Security & Access Control:** Implemented high-performance PostgreSQL-based session management, OAuth2 authentication, and fine-grained role-based access control (RBAC).
- **API-First Engineering:** Introduced automated OpenAPI/Swagger documentation pipelines with client code generation, accelerating integration between federated research nodes.
- **DevOps & Automation:** Built and optimised Docker-based CI/CD pipelines enabling isolated, reproducible deployments under agile TDD workflows.

Nov 2023 – Aug 2025 **Freelance Software Engineer, Consultant & Trainer**, *Self-employed*

- Provided technical consulting on AI-powered systems, IoT architectures and process automation for clients, including building-management software (service charges, maintenance and repairs) and a law firm.
- Delivered professional language training (German/English) for international professionals and students.

Jul 2021 – Sep 2023 **Software Engineer – IoT & Cloud**, *Beko Technologies GmbH*, Neuss

- Developed and optimised industrial IoT software solutions in embedded and cloud environments using Qt and C++.
- Designed and rolled out automated CI/CD pipelines, reducing deployment time by **30%**.
- Tuned legacy system performance by optimising data structures and SQL queries, achieving a documented **25%** efficiency gain across key data pipelines.
- Authored technical system documentation and established team-wide knowledge-sharing processes.

Dec 2020 – May 2021 **Software Engineer**, *Markt-intern Verlag GmbH*, Düsseldorf

- Extended and maintained business-critical subscription management systems.
- Systematised testing, automated debugging workflows, and refactored code to improve system stability.

Jun 2017 – Jul 2020 **Student Research Assistant – Software Development**, *IKV, RWTH Aachen University*, Aachen

- Core developer of the engineering simulation software **SphaeroSim** (C++).
- Parallelised the three main simulation algorithms with OpenMP and optimised memory usage with `std::vector` and Eigen data structures, improving simulation performance by **50%**.
- Developed data-driven models with Python and RapidMiner and implemented a simulated-annealing optimisation.
- Conducted feasibility studies for new simulator features.

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Education

2016–2020 **M.Sc. Software Systems Engineering**, *RWTH Aachen University*, Germany
Thesis: Model-Based Software Engineering and Privacy in the Smart Home

2007–2011 **B.Sc. Information Technology**, *IAUM*, Mashhad, Iran
Thesis: Customer Relationship Management (CRM Systems)

Publications

2024 Interpretable and explainable predictive machine learning models for data-driven protein engineering. *Biotechnology Advances*

2024 Peptide-based drug discovery through artificial intelligence: towards an autonomous design of therapeutic peptides. *Briefings in Bioinformatics*

2015 Virtual creativity systems for smart cities. *WASET*

2015 Trust management in e-commerce systems. *AASRJ*

2013 Security of mobile financial transactions. *IJCNCS*

Technical Skills

Programming	Python, C++, Java, JavaScript, SQL
Backend & APIs	FastAPI, RESTful APIs, OpenAPI/Swagger, Jakarta EE, OAuth2, RBAC
Databases	PostgreSQL, Neo4j (Graph Databases), MySQL, MSSQL
DevOps & Cloud	Docker, Git, GitHub Actions, CI/CD Pipelines
AI & Data Science	Explainable AI (XAI), Geometric Deep Learning, TensorFlow, HuggingFace, Jupyter
Software Eng.	Scalable Architecture, Microservices, Agile (Scrum), TDD Microservices, System Architecture
UI & Tools	React, Qt Creator, HTML5/CSS3, VS Code, IntelliJ IDEA

Languages

English	Full Professional Proficiency
German	Professional Working Proficiency
Farsi	Native Speaker

Certifications

- Deep Learning Specialization – DeepLearning.AI (Coursera)
- Microsoft Azure AI Fundamentals (AI-900)
- Microsoft Azure Fundamentals (AZ-900)

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